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Thermal Expansion Coefficients of Composite Materials Based on Energy Principles

R. A. SCHAPERY

*School of Aeronautics, Astronautics and Engineering Sciences
Purdue University, Lafayette, Indiana*

Bounds on effective thermal expansion coefficients of isotropic and anisotropic composite materials consisting of isotropic phases are derived by employing extremum principles of thermoelasticity. Inequalities between certain approximate and exact forms of the potential and complementary energy functionals are first established. These inequalities are then used in conjunction with a new method for minimizing the difference between upper and lower bounds in order to derive volumetric and linear thermal expansion coefficients. Application is made to two- and three-phase isotropic composites and a fiber-reinforced material. It is found for some important cases that the solutions are exact, and take a very simple form. We also show conditions under which the rule-of-mixtures and Turner's equation can be used for thermal expansion coefficients. Finally, simple methods for extending all results to viscoelastic composites are indicated.

INTRODUCTION

THE PROBLEM of relating effective mechanical properties of linear elastic composite materials to constituent properties has received considerable attention, as can be seen in the reviews by Hashin [1] and Pagano and Tsai [2]. As a result of this work, many practical formulas now exist for composites with either particle or fiber inclusions. Also, it has been shown by the writer that all of these results can be extended to linear viscoelastic materials in a very simple and accurate manner [3].

In contrast, little can be found on the important problem of expressing effective thermal expansion coefficients in terms of constituent properties, at least in journals in the English language. A few approximate formulas for isotropic, particulate-filled composites exist [4], with their degree of success apparently depending on the specific system studied.

Thermal Expansion Coefficients of Composite Materials

A number of articles on thermal expansion of composites have appeared recently in the Russian literature, especially in connection with fiber-reinforced materials. In an important paper by Levin [5], effective thermal expansion coefficients for two-phase anisotropic composites with isotropic phases are derived using an extension of Hill's method [6]; exact relations are given from which composite thermal coefficients may be determined if constituent properties and effective composite moduli are known. Explicit bounds on the thermal expansion coefficient of isotropic composites and unidirectional, fiber-reinforced materials with transverse isotropy are then found by using existing results for bounds on composite moduli. In a series of articles by Van Fo Fy (for example, see [7-10]), exact thermal coefficients for matrices reinforced by a doubly periodic array of hollow or solid circular, continuous fibers are developed by means of a detailed stress analysis. For a certain glass-reinforced plastic, Levin [5] gives upper bound elastic coefficients of thermal expansion which are in close agreement with the much more involved ones in [10].

In this paper we develop a new method for calculating upper and lower bounds on thermal expansion coefficients of isotropic and anisotropic composites with isotropic phases, and then give some explicit formulas for volumetric and linear coefficients of expansion of elastic and viscoelastic materials. Also, some numerical examples are presented for two and three phase systems and compared with results existing in the literature.

In contrast to the work by Levin and Van Fo Fy referenced above, composites with an arbitrary number of different constituents and arbitrary phase geometry can be treated. The method employs the complementary and potential energy principles of thermoelasticity theory in conjunction with a procedure for minimizing the difference between upper and lower bounds. It is shown that for some important cases the bounds coincide and therefore yield exact solutions. Since the prediction of effective mechanical properties has already been studied extensively, we do not consider this aspect here but simply draw upon existing solutions in the literature when they are needed.

The solutions derived here are in such a form that the basis for certain simple approximations is immediately evident. For example, simple formulas are deduced for the transverse and longitudinal thermal expansion coefficients of unidirectional, fiber-reinforced materials. It is shown that if the transverse modulus is sufficiently close to a theoretical lower bound, the rule of mixtures is an approximation to the transverse coefficient. On the other hand, a relatively high axial modulus implies that one can use a simple modulus-

modified rule of mixtures for thermal expansion in the fiber direction. Predictions of these simple formulas are shown to be in good agreement with theoretical curves in [5 and 10].

All of the elastic solutions can be extended to Laplace transformed solutions for linear viscoelastic materials by means of the correspondence principle [3]. Using this principle, Levin [5] derives the thermal expansion coefficient for an isotropic composite having a viscoelastic matrix represented by a two-element Maxwell model; usually, this model is a poor approximation of actual behavior. In the articles by Van Fo Fy, extension to more realistic viscoelastic behavior is accomplished by means of an infinite series of time operators.

In the final Section of this paper, we show how the methods in [3] can be used to easily derive realistic viscoelastic coefficients from the elastic solutions given herein. For example, it is indicated that an accurate extension can be accomplished by the quasielastic method wherein all elastic compliances (or moduli) are replaced directly by corresponding creep compliances (or relaxation moduli). This procedure yields thermal expansion coefficients for a step change in temperature, which are then substituted into the constitutive relations in [3] for use with an arbitrary temperature history.

Fiber-reinforced plastics often exhibit considerable nonlinearity in their viscoelastic response [11]. However, preliminary indications from creep and recovery measurements on such a composite are that the linear results given here can be used directly in the presence of nonlinear behavior [12]. In order to (approximately) account for the observed non-linearity, only the time-scale of viscoelastic response has to be modified to incorporate stress history.

THERMOELASTIC ENERGY PRINCIPLES

In order to derive effective thermal expansion coefficients for composite materials, we consider a specimen under a spacewise uniform temperature and in the form of a rectangular parallelepiped whose edges are parallel to coordinate axes x_i ($i = 1, 2, 3$) and with unit volume $V = 1$.^{*} This specimen is statistically homogeneous and composed of (N) phases (constituents), each of which has homogeneous mechanical and thermal properties that are different from those of any other phase; no restriction is placed on the temperature-dependence of these constituent properties.

^{*} All principal symbols are defined in the Nomenclature at the end of the text.

Maximum dimensions defining the specimen's structural inhomogeneity are assumed small compared to specimen dimensions. Further, the interaction between phases is considered to be purely mechanical and linear.

In the context of linear elastic theory, we can also assume that the specimen and phases are unstressed and unstrained when surface tractions are zero and the temperature is at some reference value, without loss of generality.

All exact and approximate solutions for effective thermal expansion coefficients will be derived by applying the minimum potential and complementary energy principles to the composite specimen. Therefore, as the first step, we shall record the pertinent energy functionals for anisotropic composites. The approach that will be followed in this paper is applicable to composites with isotropic and/or anisotropic phases; but, for algebraic simplicity, we shall study explicitly only the common case in which all phases are isotropic.

In their general form, the energy principles are usually stated with prescribed quantities being spatially varying surface displacements, tractions, temperature and body forces. However, for our purposes, it will be sufficient to let the entire surface be subjected to tractions $\hat{T}_i = \hat{\sigma}_{ij}n_j$ which are constant on each face of the specimen, omit body forces, and consider a spacewise constant temperature change T .

It is easily shown that $\hat{\sigma}_{ij}$ are equal to the volume average of actual stresses σ_{ij} existing in the composite [1]. Also, since the specimen faces are parallel to the coordinate planes, $\hat{\sigma}_{ij}$ is equal to the traction T_i on the face whose normal is in the x_j direction.

In terms of phase properties, the potential energy π for the composite specimen is

$$\pi = \int_{V=1} F_H dV - \int_S \hat{T}_i \hat{u}_i dS \quad (1)$$

where F_H is the Helmholtz free energy density [13] in terms of phase properties G , K and α :

$$F_H = G\epsilon_{ij}\epsilon_{ij} + (K/2 - G/3)\vartheta^2 - 3K\alpha T\vartheta \quad (2)$$

and the usual index notation and summation convention [13] is used.

Note that the complete form of the free energy in [13] contains an additional function of temperature which is related directly to specific heat. However, in uncoupled thermoelastic theory (i.e. the temperature is assumed to be unaffected by the strains) this function can be omitted from the phase's free energy, without loss of generality, in a stress and deformation analysis.

The complementary energy, π_c , to be used here will be defined as

the *negative* of the functional normally used. This choice of sign will simplify the notation in the equations for upper and lower bounds. We write therefore [13],

$$\pi_c = \int_{V=1} F_G dV \quad (3)$$

where F_G is the Gibb's free energy density,

$$F_G = -\frac{\sigma_{ij}\sigma_{ij}}{4G} + \left(\frac{1}{12G} - \frac{1}{18K}\right)\mathbb{H}^2 - \alpha T\mathbb{H} \quad (4)$$

A surface integral does not appear in π_c when tractions are specified over the entire surface.

If we assume that the strains in (2) and stresses in (4) are the exact distributions in the specimen, according to linear theory, it is easy to show by means of the divergence theorem that the exact energy functionals are simply related,

$$\pi_c = \pi + \frac{9}{2} T^2 \overline{K\alpha^2} \quad (5)$$

where the bar denotes a volume average,

$$\overline{K\alpha^2} \equiv \int_{V=1} K\alpha^2 dV \quad (6a)$$

in which the volume is assumed to be unity.

For homogeneous phases $n = 1, \dots, N$, each having a volume fraction v_n ,

$$\overline{K\alpha^2} = (K\alpha^2)_n v_n \equiv K_1 \alpha_1^2 v_1 + K_2 \alpha_2^2 v_2 + \dots + K_N \alpha_N^2 v_N \quad (6b)$$

where, as in (2) and (4), the repeated index implies summation over the index range.

We now write the exact complementary energy in terms of effective properties of the specimen. In this form, the composite is viewed as a homogenous, anisotropic (in general) specimen with compliances $\hat{S}_{ij}^{k\ell}$ and thermal strain coefficients $\hat{\alpha}_{ij}$, in which the stress-strain equations are

$$\hat{\epsilon}_{ij} = \hat{S}_{ij}^{k\ell} \hat{\sigma}_{k\ell} + \hat{\alpha}_{ij} T \quad (7)$$

The exact complementary energy density is

$$\pi_c = -\frac{1}{2} \hat{S}_{ij}^{k\ell} \hat{\sigma}_{ij} \hat{\sigma}_{k\ell} - \hat{\alpha}_{ij} \hat{\sigma}_{ij} T + C \quad (8)$$

where $C = C(T)$ is a function of temperature alone. Since we are

considering a composite specimen with unit volume, this expression is also the total Gibb's free energy.

It is necessary to include the additional temperature-dependent function C in (8), even though we can omit such a function in the phase modulus form (3). In the absence of surface tractions, this function is the free energy arising from internal thermal stresses that develop when the phases have different thermal expansion coefficients. Assuming zero internal stresses at $T = 0$, and recognizing that thermal stresses are proportional to T in the context of linear elasticity theory and small temperature changes, we can write $C = AT^2$, where A is constant; for large temperature changes A may depend on temperature because of the effect of temperature on properties.

Since $\hat{\alpha}_{ij} = \hat{\alpha}_{ji}$ are the components of a second order tensor, they can be calculated for any coordinate axis orientation once they are known in a particular system. Specifically, in the coordinate system $x'_i = \ell_{ij}x_j$,

$$\hat{\alpha}'_{ij} = \ell_{ik}\ell_{jl}\hat{\alpha}_{kl} \quad (9a)$$

where ℓ_{ij} is the cosine of the angle between the x'_i and x_j directions. If a specimen is orthotropic and the x_i are in the planes of symmetry, the thermal shearing coefficients are zero; i.e. $\hat{\alpha}_{12} = \hat{\alpha}_{13} = \hat{\alpha}_{23} = 0$. For this case, transformation (9a) is

$$\hat{\alpha}'_{ij} = \ell_{i1}\ell_{j1}\hat{\alpha}_{11} + \ell_{i2}\ell_{j2}\hat{\alpha}_{22} + \ell_{i3}\ell_{j3}\hat{\alpha}_{33} \quad (9b)$$

where $\hat{\alpha}_{11}$, $\hat{\alpha}_{22}$, and $\hat{\alpha}_{33}$ are linear coefficients of thermal expansion in the x_1 , and x_2 , and x_3 directions, respectively.

It is well-known that the exact value for π , equation (1), is a minimum with respect to all continuous displacements which satisfy boundary conditions on displacements; since displacements are not prescribed in the present formulation, this latter condition does not restrict the displacements we may wish to use in an approximate method. Moreover, the complementary energy, equation (3), which is the negative of the usual definition, is a maximum when the exact stresses are used; in this case the maximum is with respect to all stresses satisfying the equilibrium field equations and traction boundary conditions.

In view of these extremum properties we can use equations (5) and (8) to write

$$\pi^a_c \leq -\frac{1}{2} \hat{S}^{kl}_{ij} \hat{\sigma}_{ij} \hat{\sigma}_{kl} - \hat{\alpha}_{ij} \hat{\sigma}_{ij} T + C \leq \pi^a + \frac{9}{2} T^2 \overline{K\alpha^2} \quad (10)$$

where π^a and π^a_c are approximate values of potential and complementary energies, respectively; they are calculated by using equa-

tions (1) and (3) in which approximate displacements in (1) are continuous and approximate stresses in (3) satisfy conditions of equilibrium within the body and prescribed traction conditions on the surface.

Inequality (10) is the basic equation from which upper and lower bounds on thermal strain coefficients $\hat{\alpha}_{ij}$ will be derived. They will be expressed in terms of composite compliances, $\hat{S}_{ij}^{kl}$, and phase properties. Since methods already exist for evaluating $\hat{S}_{ij}^{kl}$ in terms of constituent properties of many technically important composites, we shall not deal with this aspect of the problem.

Explicit results will be derived from equation (10) by assuming constant strains in π^a and constant stresses in π_c^a . Although such constant approximations are normally very poor, it turns out that they can be used to derive accurate solutions for $\hat{\alpha}_{ij}$ through a proper choice for the surface tractions.

Derivation of Approximate Formulas for the Energies: As a final matter in this Section, we shall determine explicit forms for π^a and π_c^a when the pertinent mechanical variables are constant.

Constant strains to be used in π^a , equation (1), are derived from the assumed displacement distribution,

$$u_i = e_{ij}x_j \quad (11)$$

where $e_{ij} = e_{ji}$ are constants. It is seen that these six constants define a constant strain distribution since,

$$\epsilon_{ij} \equiv \frac{1}{2} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) = e_{ij} \quad (12)$$

Substitution of (11) and (12) into (1) and (2) yields

$$\pi^a = \bar{G}e_{ij}e_{ij} + \left(\frac{\bar{K}}{2} - \frac{\bar{G}}{3} \right) e^2 - 3\bar{K}\alpha Te - \hat{\sigma}_{ij}e_{ij} \quad (13)$$

where $e \equiv e_{11} + e_{22} + e_{33}$ and bars denote volume averages; for example, with homogeneous phases of volume fractions $v_1, v_2, \dots, v_N$:

$$\bar{G} \equiv G_n v_n, \quad \bar{K}\alpha \equiv (K\alpha)_n v_n \quad (14)$$

Up to this point, the strains in (13) are arbitrary. However, we shall now specify them such that they represent the "best" choice of constant strains by minimizing π^a . Thus, setting $\partial \pi^a / \partial e_{ij} = 0$ and solving for e_{ij} yields,

$$e_{ij} = \frac{\hat{\sigma}_{ij}}{2\bar{G}} - \left(\frac{1}{6\bar{G}} - \frac{1}{9\bar{K}} \right) \hat{\Theta} \delta_{ij} + \frac{\bar{K}\alpha}{\bar{K}} T \delta_{ij} \quad (15)$$

which are interpreted as stress-strain relations in terms of average properties and given stresses $\hat{\sigma}_{ij}$.

Substituting these strains back into (13) provides the final desired form,

$$\pi^a = -\frac{\hat{\sigma}_{ij}\hat{\sigma}_{ij}}{4\bar{G}} + \left(\frac{1}{12\bar{G}} - \frac{1}{18\bar{K}}\right)\hat{\mathbb{H}}^2 - \frac{\bar{K}\alpha}{\bar{K}}T\hat{\mathbb{H}} - \frac{9}{2}\frac{\bar{K}\alpha^2}{\bar{K}}T^2 \quad (16)$$

An approximate complementary energy is derived easily from (3) and (4) when we assume a constant stress distribution $\sigma_{ij} \equiv \hat{\sigma}_{ij}$,

$$\pi_c^a = -\frac{\hat{\sigma}_{ij}\hat{\sigma}_{ij}}{4G_L} + \left(\frac{1}{12G_L} - \frac{1}{18K_L}\right)\hat{\mathbb{H}}^2 - \bar{\alpha}T\hat{\mathbb{H}} \quad (17)$$

where

$$\frac{1}{G_L} \equiv \int_{V=1} \frac{dV}{G} = \frac{v_n}{G_n} \quad (18)$$

$$\frac{1}{K_L} \equiv \int_{V=1} \frac{dV}{K} = \frac{v_n}{K_n} \quad (19)$$

Note that for a homogeneous, isotropic specimen, equation (10) is identically satisfied using equality signs and energies (16) and (17).

The applied stresses $\hat{\sigma}_{ij}$ are completely arbitrary at this point. In the applications to follow, it will be seen that by choosing them in a certain manner we will be able to derive each $\hat{\alpha}_{ij}$ separately, or in various combinations, and “optimize” the upper and lower bounds.

VOLUMETRIC COEFFICIENT OF EXPANSION

The volumetric expansion coefficient $\hat{\alpha}_v$ of an isotropic *or* anisotropic composite is defined as the ratio of volume change to temperature change in the absence of surface tractions,

$$\hat{\alpha}_v \equiv \left(\frac{\hat{\mathcal{V}}}{T}\right)_{\hat{\sigma}_{ij}=0} \quad (20)$$

From equation (7) we find

$$\hat{\alpha}_v = \hat{\alpha}_{11} + \hat{\alpha}_{22} + \hat{\alpha}_{33} \quad (21)$$

Note that for isotropic composites,

$$\hat{\alpha}_{11} = \hat{\alpha}_{22} = \hat{\alpha}_{33} = \frac{\hat{\alpha}_r}{3} \quad (22)$$

In the following work it will be convenient to make use of a relation between compliances $\hat{S}_{ij}^{kl}$ and bulk modulus. The bulk modulus $\bar{K}$

for linear isotropic or anisotropic composites is defined as the ratio of applied hydrostatic stress σ to mechanical dilation,

$$\hat{K} \equiv \frac{\sigma}{(\hat{\nu} - \hat{\alpha}_v T)} \quad (23)$$

where

$$\hat{\sigma}_{ij} \equiv \sigma \delta_{ij} \quad (24)$$

and $\sigma \equiv \hat{\mathbb{H}}/3$. Substitution of equation (7) into (23) with stress state (24), yields

$$\hat{K} = \frac{1}{\hat{S}_{ii}^{jj}} \quad (25)$$

Bounds on $\hat{\alpha}_v$, equation (21), will now be found from equation (10) by applying the hydrostatic state of stress (24), and then adjusting σ so as to obtain the closest bounds possible.

Substituting stresses (24) into energies (16) and (17), and then applying inequality (10) yields

$$\begin{aligned} -\frac{\sigma^2}{2K_L} - 3\bar{\alpha}T\sigma &\leq -\frac{\sigma^2}{2\hat{K}} - \hat{\alpha}_v T\sigma + C \\ &\leq -\frac{\sigma^2}{2\bar{K}} - 3\frac{\bar{K}\bar{\alpha}}{\bar{K}}T\sigma + \frac{9}{2}T^2\left(\bar{K}\bar{\alpha}^2 - \frac{\bar{K}\bar{\alpha}}{\bar{K}}\right) \end{aligned} \quad (26)$$

Note that when $T=0$ we obtain an extension to anisotropic composites of the familiar upper (Voigt) and lower (Reuss) bounds on bulk modulus of isotropic materials [1],

$$K_L \leq \bar{K} \leq \bar{K} \quad (27)$$

It is important to observe that $\bar{K}$ and K_L arise from the separate assumptions of a constant strain distribution and a constant stress distribution, respectively, when the composite is under hydrostatic tensile stress ($\sigma > 0$) or pressure ($\sigma < 0$).

Equation (26) is now rewritten in order to isolate the unknown function $C = C(T)$,

$$\begin{aligned} -\frac{1}{2}\left(\frac{1}{K_L} - \frac{1}{\hat{K}}\right)\sigma^2 + (\hat{\alpha}_v - 3\bar{\alpha})T\sigma &\leq C \leq \frac{1}{2}\left(\frac{1}{\bar{K}} - \frac{1}{\hat{K}}\right)\sigma^2 \\ &+ (\hat{\alpha}_v - 3\frac{\bar{K}\bar{\alpha}}{\bar{K}})T\sigma + \frac{9}{2}\left(\bar{K}\bar{\alpha}^2 - \frac{\bar{K}\bar{\alpha}}{\bar{K}}\right)T^2 \end{aligned} \quad (28)$$

Consider the significance of inequality (28). The actually occurring volumetric expansion coefficient $\hat{\alpha}_v$ will be such that the left side is

never greater than the right side, regardless of the value of σ . Moreover, since C is independent of σ , we can choose σ on the left side so as to maximize that side, and choose another σ which minimizes the right side; the closer we make the left and right sides of equation (28), the closer will be the bounds on $\hat{\alpha}_v$.

Thus, by maximizing the left side and minimizing the right side we find,

$$\frac{(\hat{\alpha}_v - 3\bar{\alpha})^2}{\left(\frac{1}{K_L} - \frac{1}{\hat{K}}\right)} \leq \frac{2C}{T^2} \leq -\frac{\left(3\frac{\bar{K}\alpha}{\bar{K}} - \hat{\alpha}_v\right)^2}{\left(\frac{1}{\hat{K}} - \frac{1}{\bar{K}}\right)} + 9\left(\bar{K}\alpha^2 - \frac{\bar{K}\alpha^2}{\bar{K}}\right) \quad (29)$$

Recall that C/T^2 is finite at $T=0$ since C is proportional to T^2 for a small temperature change, T . Also, by setting $\sigma=0$ in (26) we obtain the following inequality between constituent properties,

$$\bar{K}\alpha^2 \geq \frac{\bar{K}\alpha^2}{\bar{K}} \quad (30)$$

The function C can now be bypassed by simply making use of the fact that the right side of (29) is never less than the left side, regardless of the value of C . By subtracting the right side from the left side, rearranging, and then factoring, we find

$$(\hat{\alpha}_v - \hat{\alpha}_v^{(-)}) (\hat{\alpha}_v - \hat{\alpha}_v^{(+)}) \leq 0 \quad (31)$$

where

$$\hat{\alpha}_v^{(-)} = 3\bar{\alpha} + 3\left(\frac{\bar{K}\alpha}{\bar{K}} - \bar{\alpha}\right) \frac{\left(\frac{1}{K_L} - \frac{1}{\bar{K}}\right)}{\left(\frac{1}{K_L} - \frac{1}{\bar{K}}\right)} - \Delta\hat{\alpha}_v \quad (32a)$$

$$\hat{\alpha}_v^{(+)} = 3\bar{\alpha} + 3\left(\frac{\bar{K}\alpha}{\bar{K}} - \bar{\alpha}\right) \frac{\left(\frac{1}{K_L} - \frac{1}{\bar{K}}\right)}{\left(\frac{1}{K_L} - \frac{1}{\bar{K}}\right)} + \Delta\hat{\alpha}_v \quad (32b)$$

and

$$\Delta\hat{\alpha}_v = 3 \frac{\left(\frac{1}{\bar{K}} - \frac{1}{\bar{K}}\right)^{\frac{1}{2}} \left(\frac{1}{K_L} - \frac{1}{\bar{K}}\right)^{\frac{1}{2}}}{\left(\frac{1}{K_L} - \frac{1}{\bar{K}}\right)} \left[\left(\bar{K}\alpha^2 - \frac{\bar{K}\alpha^2}{\bar{K}} \right) \left(\frac{1}{K_L} - \frac{1}{\bar{K}} \right) - \left(\frac{\bar{K}\alpha}{\bar{K}} - \bar{\alpha} \right)^2 \right]^{\frac{1}{2}} \geq 0 \quad (33)$$

Inequality (31) can be satisfied if and only if

$$\hat{\alpha}_v^{(-)} \leq \hat{\alpha}_v \leq \hat{\alpha}_v^{(+)} \quad (34)$$

and consequently $\hat{\alpha}_v^{(-)}$ is the lower bound and $\hat{\alpha}_v^{(+)}$ the upper bound on the composite volumetric expansion coefficient. Observe also that the quantity $\Delta\hat{\alpha}_v$ is positive and represents the difference between the upper (or lower) bound and the mean value $(\hat{\alpha}_v^{(-)} + \hat{\alpha}_v^{(+)})/2$; $\Delta\hat{\alpha}_v$ is, therefore, the largest possible error if the mean value is used. When $\Delta\hat{\alpha}_v = 0$, the bounds coincide, and the coefficient $\hat{\alpha}_v$ is given exactly by equation (32).

There are certain important cases for which the bounds coincide or, at least, are close to each other.

For example, $\Delta\hat{\alpha}_v \approx 0$ when $\bar{K} \approx K_L$; as mentioned earlier, this corresponds to a composite that has an approximately uniform stress distribution when it is externally pressurized, such as a fluid (or low shear modulus matrix), filled with rigid particles. For this case the "rule of mixtures" applies,

$$\hat{\alpha}_v \approx 3\bar{\alpha} \quad (35)$$

When the strain distribution is practically uniform under external pressure, the composite bulk modulus is close to the upper bound, i.e. $\bar{K} \approx \bar{K}$, and we obtain

$$\hat{\alpha}_v \approx 3 \frac{\bar{K}\alpha}{\bar{K}} \quad (36)$$

It is very interesting to observe that approximation (36) was developed heuristically and then applied by Turner [14] in a study of isotropic, metallic and plastic composites. If the constituents have nearly equal values of Poisson's ratio, equation (36) can be replaced by

$$\hat{\alpha}_v \approx 3 \frac{\bar{E}\alpha}{\bar{E}} \quad (37)$$

which was also used by Turner.

Expressions (36) and (37) were found to be in reasonable agreement with experimental data on metals and plastics in [14] and on ceramics in [15], but some of the systems were such that the difference between equations (35) and (36) or (37) was relatively small.

Clearly, whether or not these simple formulas will be accurate for a given system depends on phase geometry as well as component physical properties. For example, one might expect (36) to be more accurate than (35) for a matrix highly reinforced with randomly oriented stiff fibers; since the matrix would tend to be under the same

average strain as the fibers, a uniform strain distribution, on which (36) is based, would be more closely satisfied than a uniform stress distribution.

When all phases have equal linear thermal expansion coefficients, $\alpha_n \equiv \alpha$, equation (32) reduces to the expected exact solution $\hat{\alpha}_r = 3\alpha$.

If all phase bulk moduli are equal, equation (32) becomes indeterminate, but we can go back to equation (26). It is found that this inequality can be satisfied if and only if the solution is given exactly by the rule of mixtures, formula (35).

Application to Two-Phase Composites: All of the above results apply to the volumetric expansion coefficient of anisotropic or isotropic composites with an arbitrary number of phases. When there are only two phases, it is found, after a considerable amount of algebraic reduction, that $\Delta\hat{\alpha}_v \equiv 0$, regardless of phase properties and geometry. In fact, equation (32) yields a very simple exact solution in terms of $\hat{K}$:

$$\frac{\hat{\alpha}_v - 3\alpha_2}{3(\alpha_1 - \alpha_2)} = \frac{K_1}{\hat{K}} \frac{(K_2 - \hat{K})}{(K_2 - K_1)} \quad (38a)$$

This solution is valid for isotropic or anisotropic composites, and is identical with the result derived by Levin [5] for isotropic and cubic systems. Note that $\hat{\alpha}_v$ depends on volume fractions and phase geometry only through their effect on bulk modulus.

An alternative form of (38a) is

$$\frac{\hat{\alpha}_v}{3} = \bar{\alpha} - \frac{\frac{1}{K_L} - \frac{1}{\hat{K}}}{\frac{1}{K_1} - \frac{1}{K_2}} (\alpha_1 - \alpha_2) \quad (38b)$$

which brings out the difference between the actual expansion coefficient and the rule-of-mixtures. We see that $\hat{\alpha}_v/3 \leq \bar{\alpha}$ for the common case in which the phase with the smallest modulus has the largest thermal expansion coefficient. On the other hand, when the same phase has the smallest values of both modulus and thermal coefficient $\hat{\alpha}_v/3 \geq \bar{\alpha}$.

Observe one can use equation (38) to calculate bulk modulus, if, say, data on the composite thermal expansion coefficient and constituent properties are available. Since thermal expansion is usually easier to determine experimentally than bulk modulus, this formula may prove useful for the determination of bulk modulus.

Solution (38) is stated to be exact in the sense that it provides an exact relation between composite expansion coefficient and bulk modulus. Of course, if only upper and lower bounds on $\hat{K}$ are known in a given case, equation (38) will provide only bounds on $\hat{\alpha}_r$. When

$\alpha_1 > \alpha_2$ and $K_1 < K_2$, for example, we see from (38b) that the use of a lower bound for $\hat{K}$ yields an upper bound for $\hat{\alpha}_v$, and vice-versa.

In order to illustrate the application of (38) to a specific composite, we shall use the same example as in [4]. The example is that of a particulate-filled isotropic system with the properties:

<i>Matrix</i>	<i>Filler</i>
$\alpha_1 = 10 \times 10^{-5}/^{\circ}\text{C}$	$\alpha_2 = 1 \times 10^{-5}/^{\circ}\text{C}$
$K_1 = 7.25 \times 10^5 \text{ psi}$	$K_2 = 24.2 \times 10^5 \text{ psi}$
$G_1 = 1.55 \times 10^5 \text{ psi}$	$G_2 = 18.1 \times 10^5 \text{ psi}$

In Figure 1 we compare four different predictions of equation (38) for this composite. The rule-of-mixtures (35) and Turner's equation (36) are obtained from (38) by setting $\hat{K} = K_L$ and $\hat{K} = \bar{K}$, respectively. It was pointed out in [4] that some experimental data, while exhibiting considerable scatter, were predicted approximately by Thomas' formula $\hat{\alpha}_v = 3\alpha_1^{v_1}\alpha_2^{v_2}$; direct calculation shows that this coefficient is very close to Turner's prediction in Figure 1. Another approximation, Turner's equation (37), is shown also in [4], but it is considerably below the bulk modulus form (36).

The upper and lower bounds shown as solid lines in Figure 1 were calculated using Hashin's bounds on bulk modulus [16]. These bounds are better than those given by K_L and $\bar{K}$; they also are independent of phase geometry, but take into account constituent shear modulus. The lower bound on bulk modulus is [16]

$$\hat{K}^{(-)} = K_1 + \frac{v_2}{\frac{1}{K_2 - K_1} + \frac{v_1}{K_1 + \frac{4}{3}G_1}} \quad (39)$$

The upper bound is obtained by interchanging indices 1 and 2 everywhere. Lower bound (39) yields the upper bound on the expansion coefficient shown in Figure 1 (and vice-versa).

It is noteworthy that lower bound (39) was shown by Hashin to be an exact result for a matrix with spherical inclusions distributed in a particular way. Using a different approach, Kerner [17] assumed spherical particles and derived a bulk modulus that is identical with lower bound (39) [18]. In addition, Kerner [17] derived a thermal expansion coefficient which we have found to be identical with (38) when bulk modulus (39) is used. Therefore, the upper bound solution plotted in Figure 1 is actually Kerner's equation for thermal expansion coefficient. Because of this, we expect the upper bound on $\hat{\alpha}_v$ to be closest to the actually occurring expansion coefficient of a system having approximately spherical filler particles.

Application to a Three-Phase Composite: When a system has three or more phases we must go back to equation (32) in order to calculate bounds on the thermal expansion coefficient. As an example, consider a hypothetical composite with the following properties:

$$\begin{aligned} K_1 = K_2 = 1, K_3 = 12 \\ \alpha_1 = 10, \alpha_2 = \alpha_3 = 1 \\ v_1 = 0.5, v_2 = v_3 = 0.25. \end{aligned}$$

Units are omitted since only the ratio between corresponding phase properties are of interest here.

Using these properties we find

$$\begin{aligned} K_L \approx 1.30, \bar{K} = 3.75 \\ \bar{\alpha} = 5.50, \bar{K}\alpha/\bar{K} = 2.20 \end{aligned}$$

Also, assume $\hat{K} = 2.0$, which yields the largest difference in upper and lower bounds on $\hat{\alpha}_v$.

Substitution of these values into (32)-(34) gives

$$2.43 \leq \frac{\hat{\alpha}_v}{3} \leq 5.03 \quad (40a)$$

or, equivalently

$$\frac{\hat{\alpha}_v}{3} = 3.73 \pm 1.30 \quad (40b)$$

Note that the actual value of $\hat{\alpha}_v$ falls between the rule-of-mixtures (35) and Turner's prediction (36).

Considering the large differences in phase properties, the fact that $\hat{K}$ was chosen to obtain the largest possible spread in the bounds, and the absence of information on phase geometry and shear modulus, these bounds seem to be reasonably close. However, whether or not it is possible to improve these bounds by using different stresses $\hat{\sigma}_{ij}$ without specifying phase geometry is not known at this time.

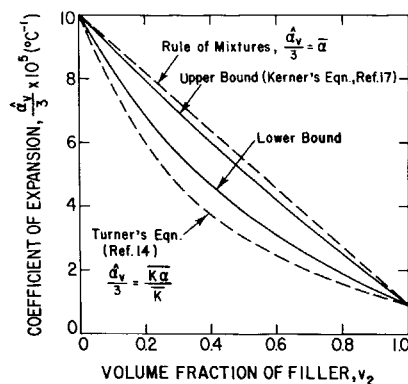


Figure 1. Coefficient of thermal expansion for an isotropic, particulate-filled system.

LINEAR COEFFICIENTS OF EXPANSION

Bounds for all coefficients $\hat{\alpha}_{ij}$ of an anisotropic composite can be found from equation (10) by using a method which is analogous to that applied in the previous section. However, here it is necessary to choose different applied stresses $\hat{\sigma}_{ij}$.

For example, if we set all $\hat{\sigma}_{ij} \equiv 0$ except for the one stress

$$\hat{\sigma}_{11} \equiv \sigma \neq 0 \quad (41)$$

bounds on $\hat{\alpha}_{11}$ will be found in terms of a single composite compliance $\hat{S}_{11}^{11}$ and phase properties. If the process is repeated for each component $\hat{\sigma}_{ij}$, bounds on all six $\hat{\alpha}_{ij}$ will be found, and in each case these bounds will be expressed in terms of a single composite compliance and phase properties. Of course, three principal coefficients, at most, are needed in order to characterize an orthotropic specimen.

Possibly closer bounds or exact solutions may be obtained by applying more than one stress component at a time, since the optimization process can be made with respect to two (or more) arbitrary stresses, rather than one. However, this improvement is made at the expense of finding each set of bounds in terms of more than one composite compliance and expansion coefficient.

We shall develop and apply equations for only $\hat{\alpha}_{11}$ using the single non-zero stress in equation (41).

Substitution of stress (41) into (16) and (17) and then into (10) yields the inequality

$$\begin{aligned} -\frac{1}{2} \frac{\sigma^2}{E_L} - \bar{\alpha} T \sigma \leq -\frac{1}{2} \frac{\sigma^2}{\hat{E}_1} - \hat{\alpha}_{11} \sigma T + C \leq -\frac{1}{2} \frac{\sigma^2}{E_U} - \frac{\bar{K}\alpha}{\bar{K}} T \sigma \\ + \frac{9}{2} T^2 \left(\bar{K}\alpha^2 - \frac{\bar{K}\alpha^2}{\bar{K}} \right) \end{aligned} \quad (42)$$

where we have introduced the following definitions

$$\hat{E}_1 \equiv 1/\hat{S}_{11}^{11} \quad (43a)$$

$$\frac{1}{E_U} \equiv \frac{1}{3} \left(\frac{1}{\bar{G}} + \frac{1}{3\bar{K}} \right); \frac{1}{E_L} \equiv \frac{1}{3} \left(\frac{1}{G_L} + \frac{1}{3K_L} \right) = \frac{v_i}{E_i} \quad (43b)$$

The last relation, $E_L^{-1} = v_i/E_i$, is derived by expressing G_L and K_L in terms of Young's modulus and Poisson's ratio of each phase.

The physical significance of $\hat{E}_1$ may be seen from equation (7). It is simply the tensile modulus under uniaxial loading along the material's x_1 -direction. The significance of E_U and E_L is found by setting $T = 0$ in equation (42); recall that $C(0) = 0$. We find

$$E_L \leq \hat{E}_1 \leq E_U \quad (44)$$

Thus, E_U and E_L are upper and lower bounds, respectively, on the uniaxial modulus; this is a well-known result for the case of isotropic composites [2, 19].

Upper and lower bounds on $\hat{\alpha}_{11}$ are now derived by applying the same technique used in the previous Section. Thus,

$$\hat{\alpha}_{11}^{(-)} \leq \hat{\alpha}_{11} \leq \hat{\alpha}_{11}^{(+)} \quad (45)$$

where

$$\hat{\alpha}_{11}^{(-)} = \bar{\alpha} + \left(\frac{\overline{K\alpha}}{\bar{K}} - \bar{\alpha} \right) \frac{\left(\frac{1}{E_L} - \frac{1}{\hat{E}_1} \right)}{\left(\frac{1}{E_L} - \frac{1}{E_U} \right)} - \Delta\hat{\alpha}_{11} \quad (46a)$$

$$\hat{\alpha}_{11}^{(+)} = \bar{\alpha} + \left(\frac{\overline{K\alpha}}{\bar{K}} - \bar{\alpha} \right) \frac{\left(\frac{1}{E_L} - \frac{1}{\hat{E}_1} \right)}{\left(\frac{1}{E_L} - \frac{1}{E_U} \right)} + \Delta\hat{\alpha}_{11} \quad (46b)$$

and

$$\Delta\hat{\alpha}_{11} = \frac{3 \left(\frac{1}{\hat{E}_1} - \frac{1}{E_U} \right)^{\frac{1}{2}} \left(\frac{1}{E_L} - \frac{1}{\hat{E}_1} \right)^{\frac{1}{2}}}{\left(\frac{1}{E_L} - \frac{1}{E_U} \right)} \left[\left(\overline{K\alpha^2} - \frac{\overline{K\alpha}^2}{\bar{K}} \right) \left(\frac{1}{E_L} - \frac{1}{E_U} \right) - \frac{1}{9} \left(\bar{\alpha} - \frac{\overline{K\alpha}}{\bar{K}} \right)^2 \right]^{\frac{1}{2}} \geq 0 \quad (47)$$

Obviously, equations (46) and (47) will provide bounds on the coefficients $\hat{\alpha}_{22}$ and $\hat{\alpha}_{33}$ by simply substituting $\hat{E}_2$ and $\hat{E}_3$, respectively, for $\hat{E}_1$. Furthermore, for an orthotropic material with coordinate axes x_i along the directions of material symmetry, $\hat{\alpha}_{12} = \hat{\alpha}_{13} = \hat{\alpha}_{23} \equiv 0$; the above equations are therefore sufficient for deriving bounds on all linear expansion coefficients. Since equation (38) for $\hat{\alpha}_r = \hat{\alpha}_{11} + \hat{\alpha}_{22} + \hat{\alpha}_{33}$ is exact for two-phase systems, it may be desirable to use it in conjunction with the above equation (46).

For composites having less than orthotropic symmetry, a shearing stress rather than a normal stress would have to be applied in order to derive the thermal shearing coefficients $\hat{\alpha}_{12}$, $\hat{\alpha}_{13}$ and $\hat{\alpha}_{23}$.

Since equations (32) and (33) are so similar to (46) and (47), most of the observations made earlier for the volumetric expansion coefficient apply here. As before, $\Delta\hat{\alpha}_{11}$ represents the largest possible error

if the mean value between upper and lower bounds is used.

For a composite in which the actual uniaxial modulus is sufficiently close or equal to the lower bound,

$$\hat{E}_1 \approx E_L \quad (48)$$

the rule of mixtures can be used

$$\hat{\alpha}_{11} \approx \bar{\alpha} \quad (49)$$

On the other hand, when

$$\hat{E}_1 \approx E_U \quad (50)$$

another simple formula results,

$$\hat{\alpha}_{11} \approx \frac{\overline{K\alpha}}{\bar{K}} \quad (51)$$

Application to a Unidirectional, Fiber-Reinforced Composite: These results lead to some simple, approximate formulas for the technically important composite consisting of a matrix with relatively stiff, parallel fibers that are continuous and straight.

Consider the axial coefficient of expansion first, in which the fibers are parallel to x_1 . Hill [6] has shown that the tensile modulus $\hat{E}_a \equiv \hat{E}_1$ in the axial direction is bounded from below by the rule-of-mixtures when all fibers have the same mechanical properties:

$$\hat{E}_a \geq \bar{E} \equiv E_1 v_1 + E_2 v_2 \quad (52)$$

where "1" refers to matrix and "2" refers to fiber. Moreover this modulus is bounded from above by E_U , equation (43b). Direct calculation of the upper bound E_U and lower bound $\bar{E}$ shows that these bounds normally are very close. This implies Turner's formula, equation (51), often will be a good approximation to the axial coefficient, $\hat{\alpha}_a \equiv \hat{\alpha}_{11}$.

When the Poisson's ratio of the matrix is equal to that of the fiber, these modulus bounds coincide, and the axial modulus is given exactly by the rule of mixtures (52); this result is actually valid with an *arbitrary* number of different fiber materials having the same Poisson's ratio, since the fibers and matrix do not interact under loading in the x_i -direction. Thus, equation (51) is *exact* and the axial coefficient can be expressed in terms of the Young's modulus of each phase:

$$\hat{\alpha}_a = \frac{\overline{E\alpha}}{\bar{E}} \equiv \frac{E_1 \alpha_1 v_1 + E_2 \alpha_2 v_2}{E_1 v_1 + E_2 v_2} \quad (53)$$

Note that the axial coefficient $\overline{E\alpha}/\bar{E}$ is exact for an arbitrary num-

ber of different kinds of fibers as long as they all have the same Poisson's ratio, with or without transverse isotropy of the composite.

It was mentioned in the introduction that some numerical results for a glass fiber-reinforced plastic are given in [5] and [10]. Figure 2 shows the axial coefficient predicted in each of these papers. The component properties used are:

Matrix	Fiber
$\alpha_1 = 6 \times 10^{-5}/^{\circ}\text{C}$	$\alpha_2 = 0.49 \times 10^{-5}/^{\circ}\text{C}$
$E_1 = 0.44 \times 10^6 \text{ psi}$	$E_2 = 9.8 \times 10^6 \text{ psi}$
$\nu_1 = 0.382$	$\nu_2 = 0.2$

The only restriction on fiber cross-section and arrangement in [5] is that the specimen must be transversely isotropic and statistically homogeneous. The exact solution in [10] is for a transversely isotropic composite having a hexagonal array of circular fibers. In both cases the fibers are solid.

The rule-of-mixtures (49) and modulus-modified rule-of-mixtures, equations (51) and (53), are shown also in Figure 2. Although the Poisson's ratios are not equal, equation (53) is seen to be very close to the solutions from [5] and [10], especially for the usual case in which $\nu_2 \geq 0.5$.

From results in [5] (for transverse isotropy) we derive the following exact relation which shows such closeness can be usually expected when $\hat{E}_a \approx \bar{E}$:

$$\hat{\alpha}_a \frac{\hat{E}_a}{\bar{E}} - \frac{\bar{E}\alpha}{\bar{E}} = \frac{\alpha_1(1 + \nu_1) - \alpha_2(1 + \nu_2)}{\nu_1 - \nu_2} \cdot \frac{\hat{E}_a - \bar{E}}{\bar{E}} \quad (54)$$

The difference $\hat{E}_a - \bar{E}$ is bound from above and below by expressions that are proportional to $(\nu_1 - \nu_2)^2$ [6], so that the term $\nu_1 - \nu_2$ in the denominator creates no problem when it is small.

It is also of interest to examine the bounds predicted by equations (46) and (47) for the particular properties given above. First of all, we find that the maximum relative spread between upper and lower

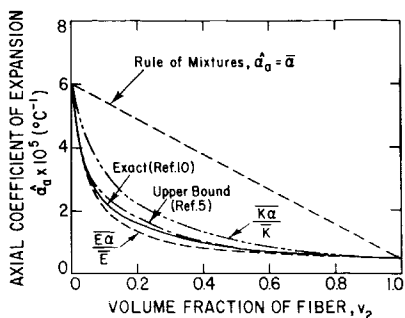


Figure 2. Axial coefficient of thermal expansion for a unidirectional glass fiber-reinforced plastic.

bounds on axial modulus, $(E_U - \bar{E})/\bar{E}$, is 3.4%, and it occurs at a fiber content of approximately $v_2 = 0.1$. This spread decreases monotonically with v_2 , and, for example, is only 0.77% when $v_2 = 0.5$. Although this difference is very small, we have found that $\Delta\hat{\alpha}_{11} \equiv \Delta\hat{\alpha}_a$, equation (47), is not always negligible. In fact, by setting $\hat{E}_a = \bar{E}$ and $v_2 = 0.5$, equations (46) and (47) yield

$$\hat{\alpha}_a \times 10^5 = 1.05 \pm 0.33 \quad (55)$$

Since the mean value, 1.05, is essentially $\overline{K\alpha}/\bar{K}$, it is seen from Figure 2 that the difference between this value and the exact solution is almost equal to $\Delta\hat{\alpha}_a$.

Consider now the prediction of the transverse thermal expansion coefficients. Transverse tensile moduli are often close to the lower bound, E_L ; hence, the mean values for $\hat{\alpha}_{22}$ and $\hat{\alpha}_{33}$ will be close to the rule-of-mixtures $\bar{\alpha}$. However, in view of the above numerical findings for a typical composite, the difference between the bounds and mean value, $\Delta\hat{\alpha}_{22}$ or $\Delta\hat{\alpha}_{33}$, may be large unless the transverse moduli are extremely close to E_L . It may be possible to improve the bounds by considering additional surface stresses and/or a plane strain formulation of the energy principles; but this has not been studied.

When all fibers have the same mechanical properties, existing exact relations can be used instead of (46) and (47). Specifically, from [7] the transverse thermal expansion coefficient in the x_2 -direction is:

$$\hat{\alpha}_{22} = \alpha_1 + (\alpha_1 - \hat{\alpha}_a) \hat{\nu}_{21} - (\alpha_1 - \alpha_2) (1 + \nu_2) \frac{\nu_1 - \hat{\nu}_{21}}{\nu_1 - \nu_2} \quad (56)$$

where $\hat{\nu}_{21}$ is Poisson's ratio under axial loading, $\hat{\sigma}_{11}$, only:

$$\hat{\nu}_{21} \equiv - \frac{\hat{\epsilon}_{22}}{\hat{\epsilon}_{11}} \quad (57)$$

A similar expression is valid for $\hat{\alpha}_{33}$, but $\hat{\nu}_{21}$ is replaced by $\hat{\nu}_{31}$:

$$\hat{\nu}_{31} \equiv - \frac{\hat{\epsilon}_{33}}{\hat{\epsilon}_{11}} \quad (58)$$

With cubic symmetry or transverse isotropy we set $\hat{\alpha}_{22} = \hat{\alpha}_{33} \equiv \hat{\alpha}_T$ and $\hat{\nu}_{21} = \hat{\nu}_{31} \equiv \hat{\nu}$; but (56) is *not* restricted to these cases.

It should be added that (56) was derived in [7] for a matrix reinforced by a doubly periodic system of hollow or solid circular fibers of various diameters. However, results in [5] can be used to show that this expression is exact, almost regardless of fiber geometry and distribution; the only limitations in [5] are statistical homogeneity and transverse isotropy. It has not yet been shown that (56) is valid for any

statistically homogeneous fiber distribution having transverse *anisotropy*.

Notice that the only effective properties of the composite appearing in (56) are $\hat{\nu}_{21}$ and $\hat{\alpha}_a$. With transverse isotropy it has been found that the rule-of-mixtures is normally a very good approximation for $\hat{\nu}_{21} = \hat{\nu}$ [2, 21, 22]; viz.,

$$\hat{\nu} \approx \bar{\nu} \equiv \nu_1 \nu_1 + \nu_2 \nu_2 \quad (59)$$

Moreover, Hill [6] has shown

$$\hat{\nu} \begin{matrix} > \\ < \end{matrix} \bar{\nu} \text{ according as } (\nu_1 - \nu_2)(k_1 - k_2) \begin{matrix} > \\ < \end{matrix} 0 \quad (60)$$

where

$$k_1 \equiv \frac{1}{2} \frac{E_1}{(1 + \nu_1)(1 - 2\nu_1)} \text{ and } k_2 \equiv \frac{1}{2} \frac{E_2}{(1 + \nu_2)(1 - 2\nu_2)} \quad (61)$$

are plane-strain bulk moduli.

Using approximation (59), we can write the transverse expansion coefficient as

$$\begin{aligned} \hat{\alpha}_T &\approx \alpha_1 + (\alpha_1 - \hat{\alpha}_a) \bar{\nu} - (\alpha_1 - \alpha_2)(1 + \nu_2) \nu_2 \\ &= (1 + \nu_1) \alpha_1 \nu_1 + (1 + \nu_2) \alpha_2 \nu_2 - \hat{\alpha}_a \bar{\nu} \end{aligned} \quad (62)$$

if $|\bar{\nu} - \hat{\nu}| / |\nu_1 - \nu_2| \ll \nu_2$.*

For the particular numerical example under consideration (62) is actually an upper bound since $\hat{\nu} \leq \bar{\nu}$ and $(\alpha_1 - \alpha_2)/(\nu_1 - \nu_2) > 0$. This transverse coefficient of expansion (62) is plotted in Figure 3 and is seen to agree well with the results from [5] and [10]; for most of the range of fiber volume fraction the axial coefficient is relatively small, so that very little error is introduced by using the approximation $\hat{\alpha}_a \approx \bar{E}\alpha/\bar{E}$.

Finally, it is of interest to observe how close this transverse expansion coefficient is to the rule-of-mixtures, as predicted by the mean value of the bounds in equation (46) when the transverse modulus is close to the lower bound E_L . The initial increase in the coefficient is apparently due to axial restraint of the fibers since the matrix

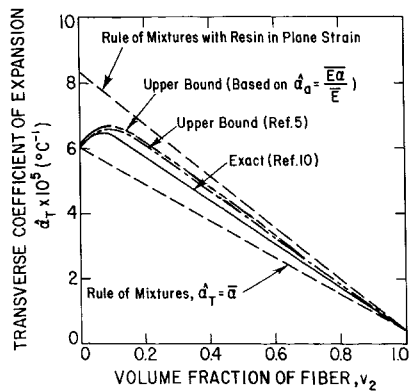


Figure 3. Transverse coefficient of thermal expansion for a unidirectional, glass fiber-reinforced plastic.

* When not valid, bounds on $\hat{\nu}$ in [6] can be used.

is practically in a state of plane strain, except at very small fiber volume fractions; the effective expansion coefficient for plane strain is $(1 + \nu_1)\alpha_1$ and the corresponding rule-of-mixtures is

$$\hat{\alpha}_T = (1 + \nu_1)\alpha_1 v_1 + \alpha_2 v_2 \quad (63)$$

which is shown also in Figure 3.

VISCOELASTIC BEHAVIOR

Extension of the results in this paper to linear viscoelastic materials is readily accomplished by using the methods in [3]. Since application was made in [3] to mechanical properties, we consider here only their application to thermal coefficients.

Viscoelastic properties are generally very sensitive to temperature. However, it is often possible to assume thermorheologically simple behavior [23] of each phase; with this assumption, all temperature-dependence is incorporated into the time-scale for viscoelastic response. If the composite exhibits this simple behavior as well, constitutive equations with transient stresses and temperature can be written as [3],

$$\epsilon_{ij} = \int_0^t \hat{S}_{ij}^{kl}(\xi - \xi') \frac{d\sigma_{kl}}{d\tau} d\tau + \int_0^t \hat{\alpha}_{ij}(\xi - \xi') \frac{dT}{d\tau} d\tau \quad (64)$$

where $\hat{S}_{ij}^{kl}(\xi)$ are creep compliances and $\hat{\alpha}_{ij}(\xi)$ define the strain response due to a unit change in temperature. Also, ξ is the so-called reduced-time defined as

$$\xi \equiv \xi(t) = \int_0^t \frac{dt'}{a_T[T(t')]}; \xi' \equiv \xi(\tau) \quad (65)$$

and a_T is a function of temperature that reflects the effect of temperature on material response.

The development and application of equation (64) was covered in [3], and will not be reported here. What is of interest now is the calculation of $\hat{\alpha}_{ij}(\xi)$ in terms of known phase properties.

Note that tensor transformation law (9) can be used to relate coefficients $\hat{\alpha}_{ij}(\xi)$ in one coordinate system to those in another. For example, with orthotropic composites it is sufficient to evaluate the principle thermal expansion coefficients $\hat{\alpha}_{11}(\xi)$, $\hat{\alpha}_{22}(\xi)$, and $\hat{\alpha}_{33}(\xi)$ in terms of phase properties.

In order to illustrate the methods, consider the calculation of the axial expansion coefficient $\hat{\alpha}_a(\xi) \equiv \hat{\alpha}_{11}(\xi)$ of a glass fiber-reinforced plastic. If we use relation (53) and assume that the glass is elastic and the plastic is viscoelastic having a relaxation modulus $E_1(\xi)$ and constant coefficient α_1 , the approximate quasi-elastic method in [3] yields

$$\hat{\alpha}_a(\xi) \simeq \frac{E_1(\xi)\alpha_1v_1 + E_2\alpha_2v_2}{E_1(\xi)v_1 + E_2v_2} \quad (66)$$

This function can be substituted directly into (64) in order to predict response to transient temperature.

In view of the basis for this quasi-elastic method, coefficient (66) is expected to be a very good approximation as long as the variation of $E_1(\xi)$ is spread over several decades of reduced-time ξ ; this is generally true for plastics.

Other methods of evaluating $\hat{\alpha}_a(\xi)$ involve application of the correspondence principle. According to this principle the exact p -multiplied Laplace transform (Carson transform) of $\hat{\alpha}_a(\xi)$ is

$$\tilde{\hat{\alpha}}_a = \frac{\tilde{E}_1\alpha_1v_1 + E_2\alpha_2v_2}{\tilde{E}_1v_1 + E_2v_2} \quad (67)$$

where

$$\tilde{\hat{\alpha}}_a \equiv p \int_0^\infty e^{-p\xi} \hat{\alpha}_a(\xi) d\xi \quad (68)$$

with a similar definition for $\tilde{E}_1$. Given $E_1(\xi)$ analytically or numerically, the right-hand side of (67) is a known function of real- p , which can be easily inverted to find $\hat{\alpha}_a(\xi)$ using the methods in [3].

If the creep compliance $D_1(\xi)$, rather than $E_1(\xi)$, is given, exact relation $\tilde{E}_1\tilde{D}_1 = 1$ can be used to find $\tilde{E}_1$. Or, more directly, $E_1(\xi)$ can often be found quite accurately by means of the quasi-elastic approximation $E_1(\xi)D_1(\xi) = 1$.

These methods are applied just as easily to all of the other elastic relations developed in this paper. In addition, bounds on elastic functions carry over to the viscoelastic functions [3]. With the quasi-elastic method, one substitutes creep or relaxation functions of each phase directly into (32) or (46), for example. Note, however, it is

necessary that all viscoelastic phases have the same a_T in order to use the resulting coefficients in (64) for transient temperatures [3].

CONCLUDING REMARKS

Bounds on thermal expansion coefficients of linear elastic and viscoelastic composite materials with temperature-dependent properties have been derived. In addition, some simple approximations for isotropic and anisotropic materials were deduced and compared with existing theory.

The method involved application of one or more boundary stresses, which were then chosen so as to minimize the difference between upper and lower bounds; although stresses were applied to the specimen in the course of deriving solutions, the resulting coefficients define thermal expansion of an unloaded specimen.

Had we formulated the problem in terms of prescribed boundary displacements, equivalent results would have been found, but in terms of coefficients in the inverse of composite stress-strain relation (7). These inverse coefficients define the average stresses developed under a temperature change when all boundaries are prevented from moving.

A possible method for extending the present results to nonlinear behavior is discussed in [12].

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NOMENCLATURE

- $\hat{E}_i$ = composite Young's modulus in x_i direction.
- E, E_n = phase Young's modulus.
- F_G = Gibb's free energy density.
- F_H = Helmholtz free energy density.
- G, G_n = phase shear modulus.
- i, j, k, ℓ = indices associated with coordinate direction 1, 2, or 3.
- K, K_n = phase bulk modulus.
- $\hat{K}$ = composite bulk modulus.
- m, n = indices designating a particular phase ($m, n = 1, 2, \dots, N$).

Thermal Expansion Coefficients of Composite Materials

n_i	= direction cosines of vector normal to surface.
S	= surface.
$\hat{S}_{ij}^{kl}$	= composite compliances.
T	= temperature change from reference value.
$\hat{T}_i$	= applied surface traction.
u_i	= displacement components.
$\hat{u}_i$	= surface displacements.
$\hat{v}_n$	= volume fraction of n th phase.
$\hat{V}$	= specimen volume = 1.
x_i	= Cartesian coordinates.
α, α_n	= phase linear coefficient of thermal expansion.
$\hat{\alpha}_r$	= composite volumetric expansion coefficient.
$\hat{\alpha}_{ij}$	= composite linear coefficients of thermal strain (or expansion).
δ_{ij}	= Kronecker delta.
ϵ_{ij}	= strain components.
$\hat{\epsilon}_{ij}$	= average strains in composite specimen.
$\textcircled{H}$	= $\sigma_{11} + \sigma_{22} + \sigma_{33}$.
ϑ	= $\epsilon_{11} + \epsilon_{22} + \epsilon_{33}$.
ν, ν_n	= phase Poisson's ratio.
$\hat{\nu}$	= composite Poisson's ratio.
ξ	= reduced time.
π, π_c	= potential and complementary energies.
σ_{ij}	= stress components.
$\hat{\sigma}_{ij}$	= applied stresses.

REFERENCES

1. Z. Hashin, "Theory of Mechanical Behavior of Heterogeneous Media," (Review Article), *Appl. Mech. Revs.*, Vol. 17 (1964), p. 1.
2. N. J. Pagano and S. W. Tsai, "Micromechanics of Composite Media," in Composite Materials Workshop, ed. by S. W. Tsai, J. C. Halpin and N. J. Pagano, Technomic Publishing Co., Inc., Stamford, Conn. (1968), p. 1.
3. R. A. Schapery, "Stress Analysis of Viscoelastic Composite Materials," *J. Comp. Mat.*, Vol. 1 (1967), p. 228.
4. L. E. Nielsen, "Mechanical Properties of Particulate-Filled Systems," *J. Comp. Mat.*, Vol. 1 (1967), p. 100.
5. V. M. Levin, "On the Coefficients of Thermal Expansion of Heterogeneous Materials," (in Russian), *Mekhanika Tverdogo Tela* (1967), p. 88.
6. R. Hill, "Theory of Mechanical Properties of Fibre-Strengthened Materials: 1. Elastic Behavior," *J. Mech. Phys. Solids*, Vol. 12 (1964), p. 199.
7. G. A. Van Fo Fy, "Elastic Constants and Thermal Expansion of Certain Bodies with Inhomogeneous Regular Structure," *Soviet Physics, Doklady*, Vol. 11 (1966), p. 176. Translation of *Doklady Akad. Nauk SSSR*, Vol. 166 (1965), p. 817.
8. G. A. Van Fo Fy and G. N. Savin, "Fundamentals of the Theory of Non-Fabric Glass-Reinforced Plastics," (in Russian), *Mekhanika Polimerov*, Vol. 1 (1965), p. 151.
9. G. A. Van Fo Fy, "Basic Relations of the Theory of Oriented Glass-Reinforced Plastics with Hollow Fibers," (in Russian), *Mekhanika Polymerov*, Vol. 2 (1966), p. 763.
10. G. A. Van Fo Fy, "Thermal Strains and Stresses in Glass Fiber-Reinforced Media," (in Russian), *Prikl. Mekh. i Teor. Fiz.* (1965), p. 101.
11. J. C. Halpin, "Introduction to Viscoelasticity," in Composite Materials Workshop,

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- ed. by S. W. Tsai, J. C. Halpin, N. J. Pagano, Technomic Publishing Co., Inc., Stamford, Conn. (1968).
12. R. A. Schapery, "On the Application of a Thermodynamic Constitutive Equation to Various Nonlinear Materials," Purdue Univ. Report No. 68-4 (June 1968).
 13. Y. C. Fung, *Fundamentals of Solid Mechanics*, Prentice-Hall, Inc., Englewood Cliffs, N. J. (1965).
 14. P. S. Turner, "Thermal Expansion Stresses in Reinforced Plastics," *J. Res. NBS*, Vol. 37 (1946), p. 239.
 15. W. D. Kingery, "Note on Thermal Expansion and Microstresses in Two-Phase Compositions," *J. Am. Ceramic Soc.*, Vol. 40 (1957), p. 351.
 16. Z. Hashin and S. Shtrikman, "A Variational Approach to the Theory of the Elastic Behavior of Multiphase Materials," *J. Mech. Phys. Solids*, Vol. 11 (1963), p. 127.
 17. E. H. Kerner, "The Elastic and Thermo-Elastic Properties of Composite Media," *Proc. Phys. Soc.*, Vol. 69B (1956), p. 808.
 18. J. E. Cuevas, "Effective Moduli Calculations of Isotropic Composites," *J. Comp. Mat.*, Vol. 2 (1968), p. 113.
 19. R. Hill, "Elastic Properties of Reinforced Solids: Some Theoretical Principles," *J. Mech. Phys. Solids*, Vol. 11 (1963), p. 357.
 20. Z. Hashin, "On Elastic Behavior of Fibre Reinforced Materials of Arbitrary Transverse Phase Geometry," *J. Mech. Phys. Solids*, Vol. 13 (1965), p. 119.
 21. Z. Hashin, "The Elastic Moduli of Fiber-Reinforced Materials," *J. Appl. Mech.*, Vol. 31 (1964), p. 223.
 22. J. M. Whitney and M. B. Riley, "Elastic Properties of Fiber Reinforced Composite Materials, *AIAA J.*, Vol. 4 (1966), p. 1537.
 23. F. Schwarzl and A. J. Staverman, "Time-Temperature Dependence of Linear Viscoelastic Behavior," *J. Appl. Phys.*, Vol. 23 (1952), p. 838.

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